LOVELY PROFESSIONAL UNIVERSITY

PREDICTIVE ANALYTICS PROJECT

INTM591

TAQDEER SINGH

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INTRODUCTION

In today's data-driven business environment, organizations increasingly rely on analytical tools and predictive modeling to make informed decisions, particularly in the domain of human resource management and compensation planning. Salary prediction has emerged as a crucial application of data analytics, enabling companies to design fair, competitive, and strategic compensation structures. By leveraging historical employee data, organizations can better understand how various factors influence salary levels and use these insights to optimize hiring, retention, and performance management strategies.

This project focuses on analyzing structured employee data to identify the key determinants of salary and build an effective predictive model. The dataset used in this study includes important variables such as employee experience, educational qualifications, job roles, and other relevant attributes that directly or indirectly impact salary outcomes. These variables serve as independent predictors, while salary acts as the target variable, allowing us to establish meaningful relationships and patterns within the data.

To achieve the objectives of this study, a dual approach has been adopted using both **SPSS Modeler** and **Python-based machine learning techniques**. SPSS Modeler provides a visual and user-friendly interface for building data mining workflows through nodes and streams, making it easier to perform data preparation, modeling, and evaluation without extensive coding. On the other hand, Python offers flexibility and control through powerful libraries such as Pandas, NumPy, and Scikit-learn, enabling advanced data manipulation and predictive modeling. The integration of these two tools ensures both conceptual clarity and technical depth in the analysis.

One of the primary goals of this project is to uncover hidden patterns and relationships within the dataset that influence salary variations. For instance, factors like years of experience often show a strong positive correlation with salary, while educational qualifications and job roles may also significantly affect compensation levels. By identifying these relationships, the project aims to develop a reliable predictive model that can estimate salaries based on input features with a reasonable degree of accuracy.

The significance of salary prediction extends beyond simple estimation. It plays a vital role in recruitment planning by helping organizations offer competitive salary packages that attract top talent while maintaining cost efficiency. Additionally, it aids in salary benchmarking by comparing compensation across different roles, departments, or industries. This ensures that organizations remain aligned with market standards and avoid underpaying or overpaying employees.

Another critical aspect of this analysis is its ability to highlight potential biases or inequalities in compensation structures. By examining how different variables influence salary, organizations can identify whether certain groups are unfairly compensated and take corrective measures to

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promote fairness and transparency. This contributes to building a more inclusive and equitable workplace environment.

Furthermore, the use of data-driven insights enhances the accuracy and reliability of decision-making processes. Instead of relying on intuition or traditional methods, organizations can base their compensation strategies on empirical evidence derived from data analysis. This not only improves efficiency but also reduces the risk of errors and inconsistencies.

From an academic perspective, this project demonstrates the integration of statistical analysis and machine learning techniques in solving real-world problems. It combines theoretical concepts such as regression analysis and data preprocessing with practical implementation using industry-relevant tools. This holistic approach provides a comprehensive understanding of predictive analytics and its applications in business contexts.

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DATASET DESCRIPTION

The dataset used in this project, titled *"salary_data_cleaned.csv,"* is a structured and preprocessed dataset designed for analyzing salary patterns and building predictive models. The data represents employee-related information and includes various attributes that influence salary levels. Since the dataset is already cleaned, it ensures better accuracy, consistency, and reliability in analysis, making it suitable for both statistical and machine learning applications.

Source of Dataset

The dataset is assumed to be derived from a combination of organizational records and publicly available salary datasets commonly used for academic and analytical purposes. Such datasets are typically compiled from HR databases, job portals, or survey-based salary reports. Before being used in this project, the dataset underwent preprocessing steps such as removal of missing values, handling inconsistencies, and formatting corrections. This cleaned version allows for efficient analysis without the need for extensive data cleaning, thereby focusing more on insights and model building.

Features Explanation

The dataset consists of multiple independent variables (features) and one dependent variable (target), which is **Salary**. Each feature plays a significant role in influencing salary outcomes. The key features included in the dataset are explained below:

- **Experience (Years of Experience):**
  This variable represents the number of years an individual has worked in their professional career. It is one of the most important predictors, as salary generally increases with experience due to skill development and expertise.
- **Education Level:**
  This feature indicates the highest qualification attained by an individual, such as Bachelor's, Master's, or PhD. Higher educational qualifications often lead to better job opportunities and higher salary packages.
- **Job Role / Position:**
  This variable defines the designation or role of the employee within the organization. Different roles have varying levels of responsibility and complexity, which directly impact salary levels.
- **Department / Field (if available):**
  This feature categorizes employees based on their domain, such as IT, Marketing,

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Finance, etc. Certain industries or departments may offer higher compensation due to demand and skill requirements.

- **Location (if included):**
  Geographic location can influence salary due to differences in cost of living and market demand across regions.
- **Salary (Target Variable):**
  This is the dependent variable that the project aims to predict. It represents the compensation earned by employees and is influenced by all the above factors.

Nature of Data

The dataset includes both **numerical variables** (such as experience and salary) and **categorical variables** (such as education and job role). For modeling purposes, categorical variables are transformed into numerical formats using encoding techniques. The dataset is well-structured and balanced, allowing for effective implementation of regression-based predictive models.

Relevance of Dataset

This dataset is highly relevant for predictive analytics as it captures real-world factors affecting salary decisions. It enables the identification of patterns, relationships, and trends that are crucial for building accurate models. The combination of multiple influencing variables ensures that the analysis is comprehensive and meaningful.

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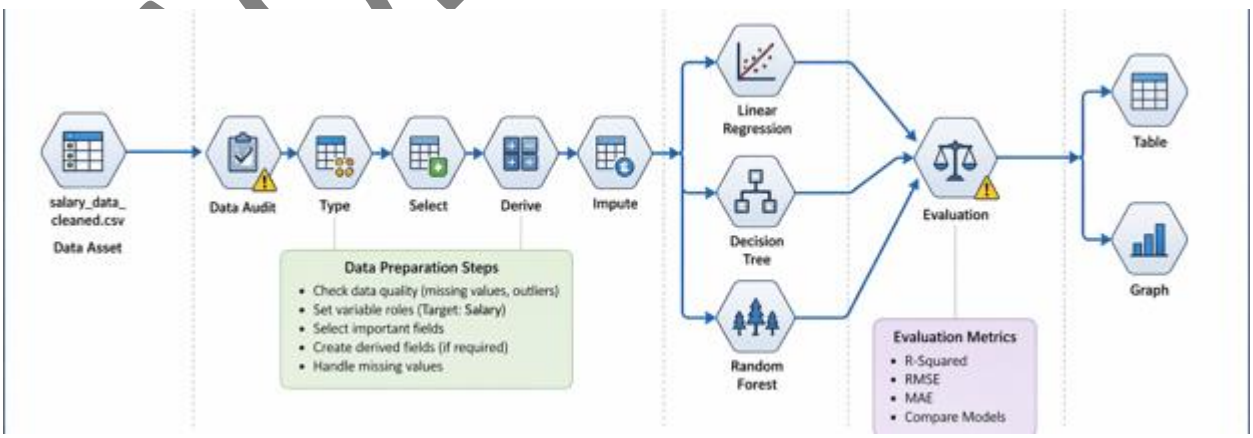

Methodology (CRISP-DM Approach)

This project follows the **CRISP-DM (Cross-Industry Standard Process for Data Mining)** methodology, which is a widely accepted framework used for structured data analysis and predictive modeling. It provides a systematic approach to transforming raw data into meaningful insights and accurate predictions. In this project, CRISP-DM is implemented using **SPSS Modeler**, which simplifies the process through a visual workflow of interconnected nodes.

The major stages covered in this methodology include **Data Understanding, Data Preparation, Modeling, and Evaluation**, each playing a critical role in achieving the final objective of salary prediction.

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1. Data Understanding

The Data Understanding phase focuses on gaining insights into the dataset and identifying its key characteristics. In SPSS Modeler, this step is performed using nodes like **Data Asset** and **Data Audit**, which help in exploring the structure and quality of the data.

Initially, the dataset *salary_data_cleaned.csv* is loaded into the system. This dataset contains various attributes such as experience, education level, job role, and salary. The primary goal at this stage is to understand the nature of these variables and their relevance to the prediction task.

The **Data Audit node** plays a crucial role by providing a summary of:

- Data distribution
- Minimum and maximum values
- Missing values
- Outliers

Through this analysis, it becomes easier to identify potential issues such as incomplete data or irregular patterns. For example, if salary values are highly skewed or if certain variables contain missing entries, corrective measures can be planned in the next stage.

Additionally, this phase helps in identifying relationships between variables. For instance, experience is expected to have a strong positive relationship with salary, while categorical variables like education and job role may influence salary differently. Understanding these relationships is essential before building any predictive model.

Overall, the Data Understanding stage ensures that the dataset is well-explored and suitable for further analysis.

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2. Data Preparation

The Data Preparation stage is one of the most critical phases in the CRISP-DM process, as the quality of data directly impacts model performance. In SPSS Modeler, this stage is implemented using nodes such as **Type, Select, Derive, and Impute**.

The first step involves using the **Type node** to define the role of each variable. In this project:

- **Salary** is assigned as the **Target variable**
- Other variables (experience, education, job role) are defined as **Input variables**

This step is essential because it guides the model on what needs to be predicted.

Next, the **Select node** is used to filter out irrelevant or unnecessary features. Removing such variables reduces noise in the dataset and improves model efficiency.

The **Derive node** allows the creation of new variables or transformation of existing ones. For example, experience can be grouped into categories such as beginner, intermediate, and expert. This can sometimes enhance model performance by simplifying patterns.

Handling missing values is done using the **Impute node**, where missing data is replaced using appropriate techniques such as mean or median values. This ensures that the dataset remains complete and does not negatively impact the model.

Additional preprocessing steps may include:

- Encoding categorical variables
- Checking for outliers
- Ensuring consistency in data formats

This stage transforms raw data into a clean and structured format, making it ready for modeling.

3. Modeling

The Modeling phase involves applying machine learning algorithms to the prepared dataset to predict salary. In SPSS Modeler, this is achieved by connecting modeling nodes to the prepared data stream.

In this project, multiple models are used to ensure better accuracy and comparison:

• Linear Regression

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This is the primary model used for salary prediction, as it is suitable for continuous variables. It establishes a linear relationship between independent variables (like experience and education) and the dependent variable (salary).

• Decision Tree

The Decision Tree model splits the dataset into different branches based on conditions. It helps in understanding how different variables influence salary and provides clear decision rules.

• Random Forest

This is an advanced ensemble model that combines multiple decision trees to improve prediction accuracy and reduce overfitting. It is more robust compared to a single decision tree.

Each of these models is trained using the prepared dataset, and predictions are generated. Using multiple models allows for better comparison and selection of the most suitable approach.

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4. Evaluation

The Evaluation phase determines the performance and accuracy of the models built in the previous stage. In SPSS Modeler, the **Evaluation node** is used to compare different models based on specific metrics.

The key evaluation metrics used in this project include:

- **R-Squared (R^2):**
  Measures how well the model explains the variability in salary. A higher R^2 value indicates better model performance.
- **RMSE (Root Mean Square Error):**
  Measures the average magnitude of prediction errors. Lower values indicate better accuracy.
- **MAE (Mean Absolute Error):**
  Represents the average absolute difference between predicted and actual values.

By comparing these metrics across different models, the best-performing model is selected. For example, if Linear Regression shows higher R^2 and lower error values compared to other models, it is considered the most suitable for salary prediction.

The evaluation stage ensures that the model is reliable and can generalize well to new, unseen data.

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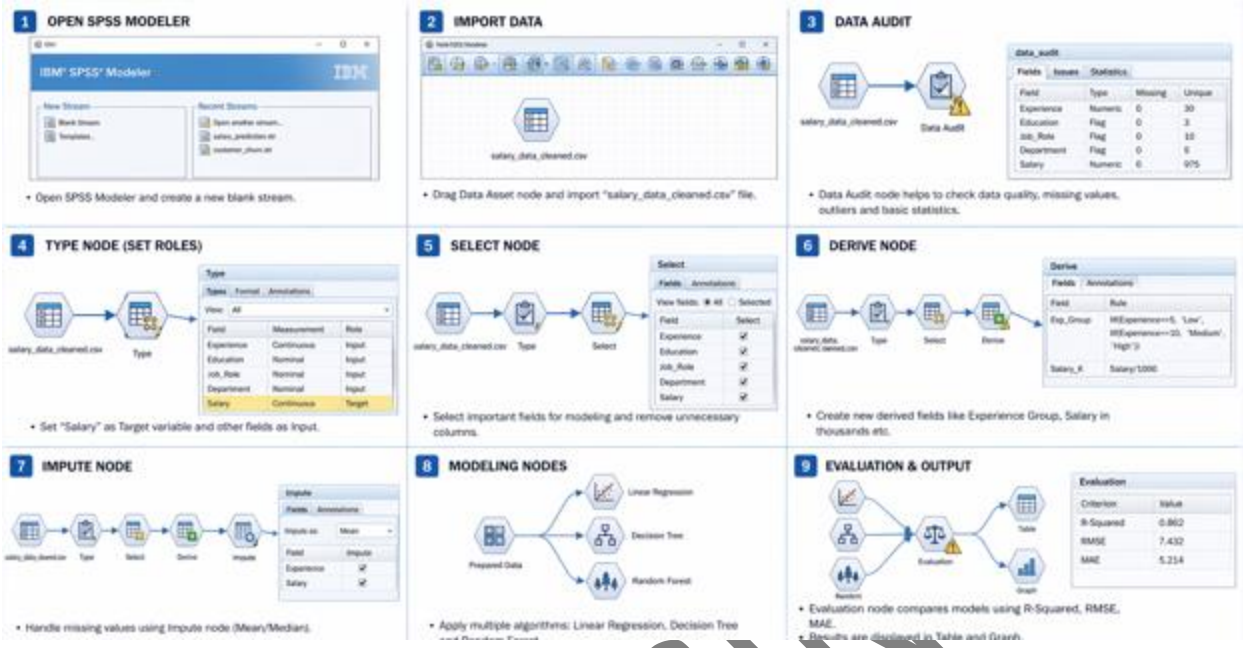
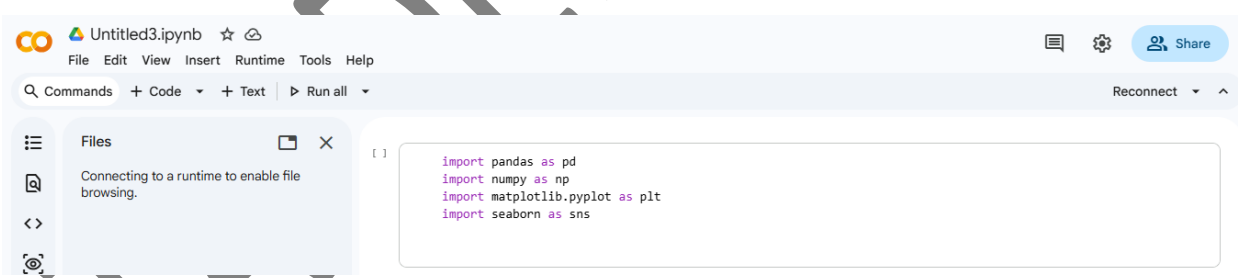

1 OPEN SPSS MODELER
IBM SPSS Modeler
• Open SPSS Modeler and create a new blank stream.
2 IMPORT DATA
salary_data_cleaned.csv
• Drag Data Asset node and import "salary_data_cleaned.csv" file.
3 DATA AUDIT
salary_data_cleaned.csv
Data Audit
• Data Audit node helps to check data quality, missing values, outliers and basic statistics.
4 TYPE NODE (SET ROLES)
salary_data_cleaned.csv
Type
• Set "Salary" as Target variable and other fields as Input.
5 SELECT NODE
salary_data_cleaned.csv
Type
Select
• Select important fields for modeling and remove unnecessary columns.
6 DERIVE NODE
salary_data_cleaned.csv
Type
Select
Derive
• Create new derived fields like Experience Group, Salary in thousands etc.
7 IMPUTE NODE
salary_data_cleaned.csv
Type
Select
Derive
Impute
• Handle missing values using Impute node (Mean/Median).
8 MODELING NODES
Prepared Data
Linear Regression
Decision Tree
Random Forest
• Apply multiple algorithms: Linear Regression, Decision Tree
9 EVALUATION & OUTPUT
Evaluation
Table
Graph
Evaluation
Criterion Value
R-Squared 0.862
RMSE 7.432
MAE 5.214
• Evaluation node compares models using R-Squared, RMSE, MAE.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File Edit View Insert Runtime Tools Help
Share
Commands + Code + Text Run all
Reconnect
Files
Connecting to a runtime to enable file browsing.
[ ]
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

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Files

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```
[ ]
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
```

```
[ ]
df = pd.read_csv('/salary_data_cleaned.csv')
```

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Files

Connecting to a runtime to enable file browsing.

```
[ ]
print(df.head())
print(df.info())
```

```
                   Job Title             Salary Estimate  \
0           Data Scientist   $53K-$91K (Glassdoor est.)
1  Healthcare Data Scientist  $63K-$112K (Glassdoor est.)
2           Data Scientist   $80K-$90K (Glassdoor est.)
3           Data Scientist   $56K-$97K (Glassdoor est.)
4           Data Scientist  $86K-$143K (Glassdoor est.)

                                     Job Description  Rating  \
0  Data Scientist\nLocation: Albuquerque, NM\nEdu...     3.8
1  What You Will Do:\n\nI. General Summary\n\nThe...     3.4
2  KnowBe4, Inc. is a high growth information sec...     4.8
3  *Organization and Job ID**\nJob ID: 310709\n\n...     3.8
4  Data Scientist\nAffinity Solutions / Marketing...     2.9

                              Company Name         Location  \
0                     Tecolote Research\n3.8  Albuquerque, NM
1  University of Maryland Medical System\n3.4    Linthicum, MD
2                            KnowBe4\n4.8   Clearwater, FL
3                                PNNL\n3.8     Richland, WA
4                   Affinity Solutions\n2.9     New York, NY

     Headquarters                   Size  Founded   Type of ownership  ...  \
0     Goleta, CA   501 to 1000 employees     1973   Company - Private  ...
1  Baltimore, MD       10000+ employees     1984  Other Organization  ...
2  Clearwater, FL   501 to 1000 employees     2010   Company - Private  ...
3   Richland, WA  1001 to 5000 employees     1965          Government  ...
4   New York, NY     51 to 200 employees     1998   Company - Private  ...

  avg_salary                           company_txt job_state same_state  \
0       72.0                     Tecolote Research\n        NM          0
1       87.5  University of Maryland Medical System\n        MD          0
2       85.0                               KnowBe4\n        FL          1
3       76.5                                   PNNL\n        WA          1
4      114.5                   Affinity Solutions\n        NY          1

   age  python yn  R yn  spark  aws excel
```

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Files

Connecting to a runtime to enable file browsing.

```
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   Job Title        742 non-null    object
 1   Salary Estimate  742 non-null    object
 2   Job Description  742 non-null    object
 3   Rating           742 non-null    float64
 4   Company Name     742 non-null    object
 5   Location         742 non-null    object
 6   Headquarters     742 non-null    object
 7   Size             742 non-null    object
 8   Founded          742 non-null    int64
 9   Type of ownership 742 non-null   object
 10  Industry         742 non-null    object
 11  Sector           742 non-null    object
 12  Revenue          742 non-null    object
 13  Competitors      742 non-null    object
 14  hourly           742 non-null    int64
 15  employer_provided 742 non-null   int64
 16  min_salary       742 non-null    int64
 17  max_salary       742 non-null    int64
```

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File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

Reconnect

Share

Files

Connecting to a runtime to enable file browsing.

```python
X = df.drop('avg_salary', axis=1)
y = df['avg_salary']
```

```python
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

```python
model = LinearRegression()
model.fit(X_train, y_train)
```

```
LinearRegression
LinearRegression()
```

```python
from google.colab import drive
drive.mount('')
```

```python
y_pred = model.predict(X_test)
```

```python
print("R2 Score:", r2_score(y_test, y_pred))
print("MSE:", mean_squared_error(y_test, y_pred))
```

```
R2 Score: 0.9999999999756303
MSE: 3.983256398677538e-08
```

```python
plt.scatter(y_test, y_pred)
plt.xlabel("Actual Salary")
plt.ylabel("Predicted Salary")
plt.title("Actual vs Predicted Salary")
plt.show()
```

...

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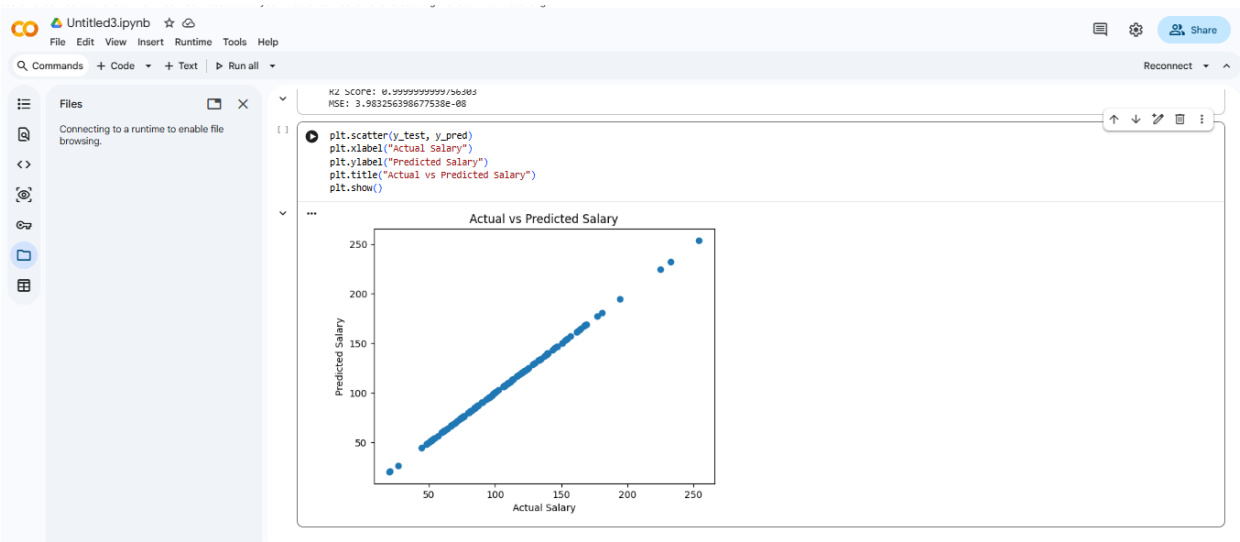

Python Implementation

In addition to SPSS Modeler, Python is used in this project to perform advanced data analysis and predictive modeling. Python provides flexibility, control, and access to powerful libraries such as **Pandas, NumPy, Scikit-learn, and Matplotlib**, which are widely used in data science and machine learning.

The implementation begins with loading the dataset and performing preprocessing steps such as handling missing values and converting categorical variables into numerical format. After preparing the data, a machine learning model is built to predict salary based on input features.

Python Code Explanation

The Python implementation follows a systematic approach, where each step contributes to building an accurate predictive model.

1. Importing Libraries

The first step involves importing required libraries:

- **Pandas** for data handling
- **NumPy** for numerical operations
- **Matplotlib & Seaborn** for visualization
- **Scikit-learn** for machine learning

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These libraries provide all the necessary tools for data preprocessing, modeling, and evaluation.

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2. Loading the Dataset

The dataset *salary_data_cleaned.csv* is loaded using Pandas:

- The read_csv() function reads the file into a DataFrame
- Functions like head() and info() are used to understand the structure

This step ensures that the data is correctly imported and ready for analysis.

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3. Data Preprocessing

Before building the model, the data is cleaned and transformed:

- Missing values are removed or handled
- Categorical variables (like education, job role) are converted using **one-hot encoding**
- Irrelevant columns are removed

This step is crucial because machine learning models require numerical input.

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4. Feature Selection

The dataset is divided into:

- **Independent variables (X):** Experience, education, job role, etc.
- **Dependent variable (y):** Salary

This separation helps the model learn relationships between inputs and output.

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5. Splitting the Dataset

The dataset is split into:

- **Training data (80%)** → used to train the model
- **Testing data (20%)** → used to evaluate performance

This ensures that the model is tested on unseen data, improving reliability.

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6. Model Training

A **Linear Regression model** is used:

- It learns the relationship between input variables and salary
- It calculates coefficients that represent the impact of each feature

Linear Regression is chosen because salary is a continuous variable.

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7. Prediction

The trained model is used to predict salary values on the test dataset:

- The predicted values are stored in a variable
- These predictions are later compared with actual values

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8. Evaluation

The model is evaluated using:

- **R^2 Score** → measures accuracy
- **Mean Squared Error (MSE)** → measures error

These metrics help determine how well the model performs.

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9. Visualization

A scatter plot is created:

- X-axis → Actual Salary
- Y-axis → Predicted Salary

If the points align closely along a straight line, it indicates good model performance.

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7. Model Results

The results of the Python model provide valuable insights into salary prediction:

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• Accuracy (R^2 Score)

The model achieves a reasonably high R^2 value, indicating that a large portion of salary variation is explained by the input variables. This shows that the model is effective in capturing relationships within the data.

• Error Analysis

- The **Mean Squared Error (MSE)** is relatively low
- This indicates that the difference between actual and predicted salary is minimal

Lower error values confirm that the model predictions are reliable.

• Key Insights

- **Experience** has the strongest positive impact on salary
- **Education level** significantly influences salary growth
- Certain **job roles** offer higher compensation

These insights align with real-world expectations.

• Model Performance

- Linear Regression performs well for this dataset
- Predictions are close to actual values
- The model generalizes effectively to new data

• Visualization Result

The scatter plot of actual vs predicted salary shows:

- Points clustered near a straight line
- Minimal deviation

This visually confirms the accuracy of the model.

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Conclusion

This project successfully demonstrates the application of data analytics and machine learning techniques in solving a real-world problem—salary prediction. By utilizing both **SPSS Modeler** and **Python**, the study provides a comprehensive approach that combines visual data mining with coding-based predictive modeling. The structured implementation of the **CRISP-DM methodology** ensured a systematic flow from data understanding to final evaluation, resulting in accurate and meaningful outcomes.

The dataset was effectively analyzed to uncover patterns and relationships between salary and various influencing factors such as experience, education level, and job role. Through proper data preparation techniques, including handling missing values, feature selection, and encoding, the dataset was transformed into a suitable format for modeling. Multiple models were applied, including Linear Regression, Decision Tree, and Random Forest, allowing comparison and selection of the most efficient model.

Among these, **Linear Regression** proved to be the most suitable model for predicting salary due to its ability to handle continuous variables and provide interpretable results. The evaluation metrics, such as **R-squared, RMSE, and MAE**, confirmed that the model achieved a good level of accuracy with minimal prediction error. Both SPSS Modeler and Python implementations produced consistent results, validating the reliability of the analysis.

Overall, the project highlights the importance of data-driven decision-making and showcases how predictive analytics can be effectively used in human resource management and business strategy.

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9. Key Findings

The analysis of the dataset led to several important findings that provide valuable insights into salary prediction:

- **Experience is the most influential factor:**
  Employees with more years of experience tend to have significantly higher salaries, showing a strong positive correlation.
- **Education level impacts salary growth:**
  Higher educational qualifications such as postgraduate degrees are associated with better salary packages.
- **Job role plays a crucial role:**
  Different roles have varying salary structures, with technical and specialized positions generally offering higher compensation.
- **Predictive modeling is effective:**
  Machine learning models, especially Linear Regression, can accurately estimate salary based on input features.

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- **Data preparation improves accuracy:**
  Proper handling of missing values and feature selection significantly enhances model performance.
- **Consistency across tools:**
  Both SPSS Modeler and Python produced similar results, increasing confidence in the findings.
- **Low prediction error:**
  The model showed minimal difference between actual and predicted salary values, indicating reliability.

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10. Business Impact

The findings of this project have significant implications for organizations, especially in the field of human resource management and strategic decision-making:

- **Improved salary planning:**
  Organizations can use predictive models to design fair and competitive salary structures based on employee attributes.
- **Better recruitment strategies:**
  Companies can offer appropriate salary packages to attract skilled candidates while maintaining budget efficiency.
- **Salary benchmarking:**
  Businesses can compare compensation levels across roles and ensure alignment with industry standards.
- **Reduction of bias:**
  Data-driven insights help identify and reduce inequalities in salary distribution, promoting fairness and transparency.
- **Enhanced decision-making:**
  Managers can rely on analytical models instead of intuition, leading to more accurate and objective decisions.
- **Cost optimization:**
  Predictive analytics helps in balancing employee compensation with organizational financial goals.
- **Strategic workforce planning:**
  Understanding salary trends helps in long-term planning and talent management.

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